



RAMAIAH
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OF APPLIED SCIENCES

Final Project Presentation

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Drone for Real-Time Image Processing

Design And Development of AI-Assisted Drone for Disaster Management

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Outline

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Title and Aim

Title : Design And Development of AI-Assisted Drone for Disaster Management

Aim:

Develop an AI-assisted of **real-time human detection** and **gesture analysis** to assist disaster relief operations.

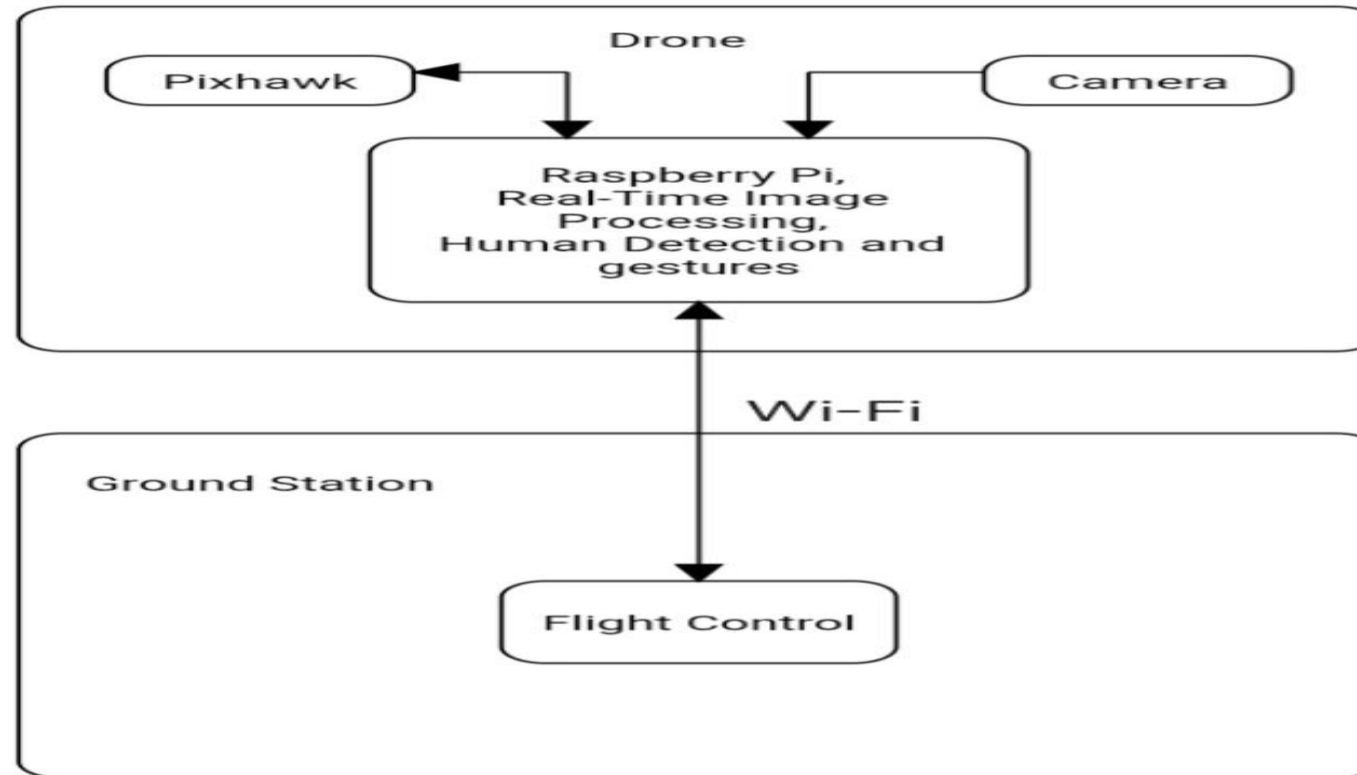
Introduction

- **Drones in Disaster Management:** Provide rapid aerial support for search and rescue in inaccessible or hazardous areas.
- **AI Integration:** Incorporating AI (YOLO, deep learning) enables **real-time human detection and gesture recognition**, reducing the cognitive load on operators.
- **Project Focus:** Design a drone system that autonomously detects people and recognizes distress gestures (e.g., hands raised) to alert rescue teams immediately.

Introduction



Block Diagram



Objectives

1. To carry out the literature review for human detection , Gesture Detection and Image processing algorithms (Yolov8 etc).
2. To design and develop a drone platform integrated with Pixhawk flight controller and flysky with the Raspberry Pi 5 (8GB).
3. To Implement image processing techniques using OpenCV and YOLOv8 to detect humans
4. To Establish communication between the Raspberry Pi and Pixhawk through the MAV Link protocol for Synchronized flight control.
5. To Test and validate the system's performance in real flight conditions to evaluate stability, accuracy, and responsiveness.

Hardware Requirements

- **Flight Controller:** Pixhawk 2.4.8 for stable flight control.
- **Compute Unit:** Raspberry Pi 5 (16GB) running Linux as the onboard computer.
- **Camera:** Arducam IMX519 (16MP), 80 FPS @720p.
- **Power & Frame:** Quadcopter frame (e.g. F450), Li-Po battery (11.1V, 5200 mAh), ESCs, brushless motors capable of ~3 kg payload.
- **Sensors:** GPS module for positioning, and RC transceiver (e.g., FlySky).

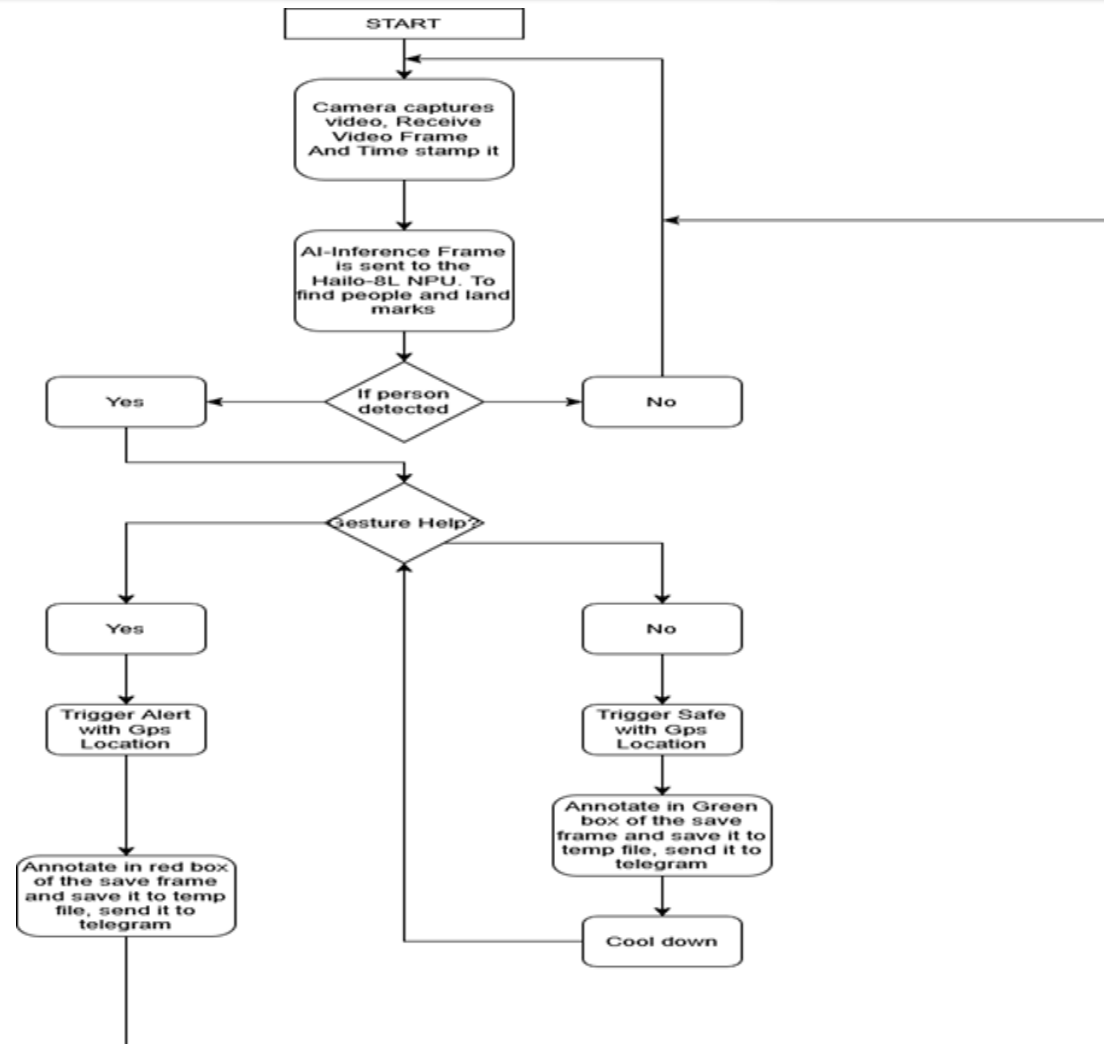
Software Requirements

- **Flight Software:** ArduPilot (Copter firmware) on Pixhawk; Ground Control: Mission Planner.
- **Languages:** Coding in Python 3.
- **AI Libraries:** OpenCV for image handling; YOLOv8 for detection/models.
- **Networking:** MAVLink protocol for commands; Wi-Fi/4G or telemetry for data; optionally Telegram Bot API for alert messages.

Hardware Architecture

- **Compute:** Raspberry Pi acts as the main controller (runs detection and logic).
- **AI Inference:** NPU (or Coral TPU) processes the YOLO model at high FPS with low CPU load.
- **Imaging:** Arducam IMX519 provides continuous video (720p@80fps) to the compute module.
- **Control:** Pixhawk handles flight stabilization; communicates with Pi via UART (MAVLink).
- **Connectivity:** Drone connects over Wi-Fi or cellular to ground station; uses Telegram API for instant alerts.

Flow Chart



Software Stack

- **Programming:** Python 3.11+ for main pipeline; multithreading to keep video processing non-blocking.
- **Computer Vision:** YOLOv8 model (COCO-Pose) to detect person and 17 body keypoints.
- **Graphics & Video:** OpenCV for drawing skeleton, boxes, and handling camera frames.
- **Streaming:** Low-latency video streaming from camera to AI accelerator.
- **Networking:** Requests/threading libraries for HTTP (Telegram API) and socket communication.

Methodology

Data Capture: Camera streams NV12 frames (720p @80 FPS).

Inference: Each frame sent to YOLOv8 pose model on the AI module.

Keypoint Analysis: Extract 17 joints; compute relative Y-coordinates of wrists, shoulders, hips.

Gesture Decision:

Hands-Up (High Alert): Both wrists above corresponding shoulders $\Rightarrow$ **Distress**.

Hands-Down (Safe): Both wrists below corresponding hips $\Rightarrow$ **Safe**.

Require gesture to be stable >1s to avoid false triggers.

Action: If *Distress*, overlay alert visuals and initiate alert protocol. Otherwise, mark *Safe*

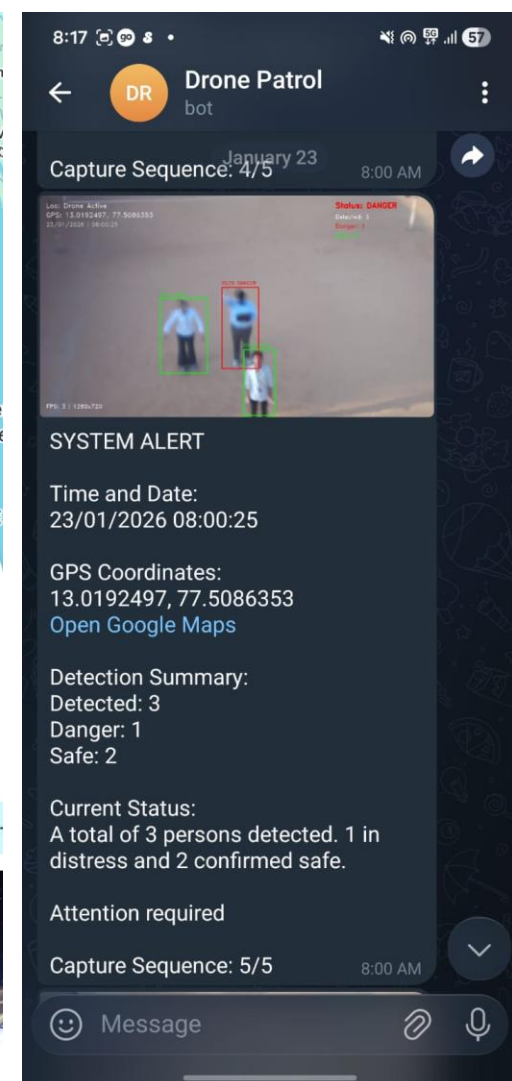
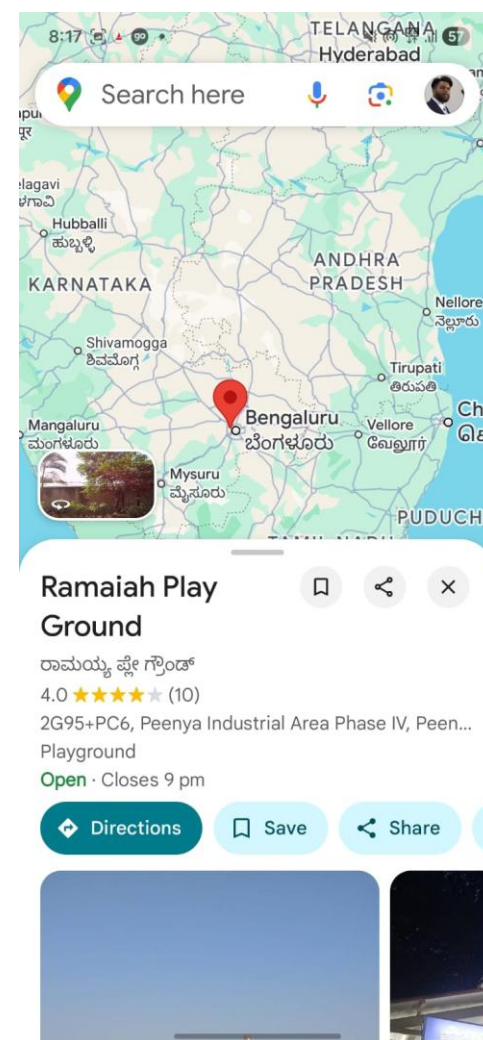
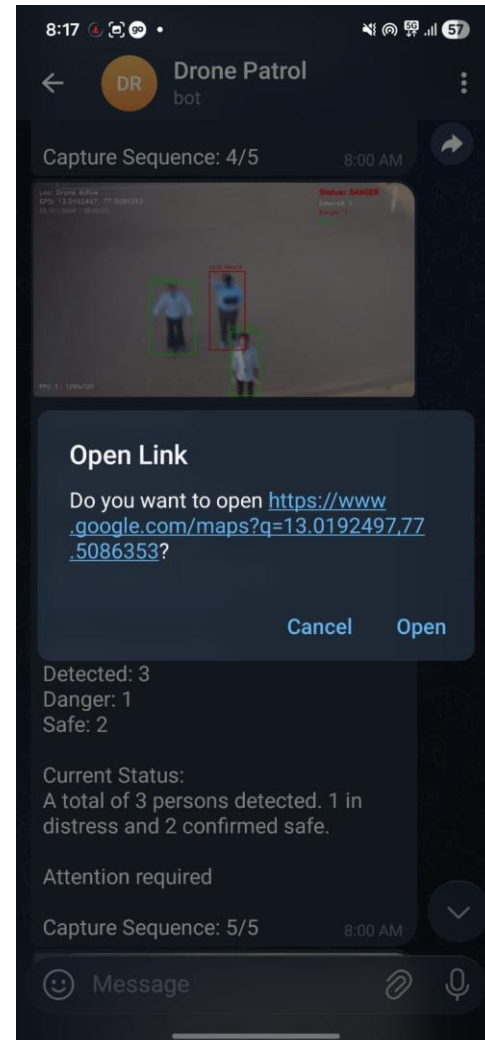
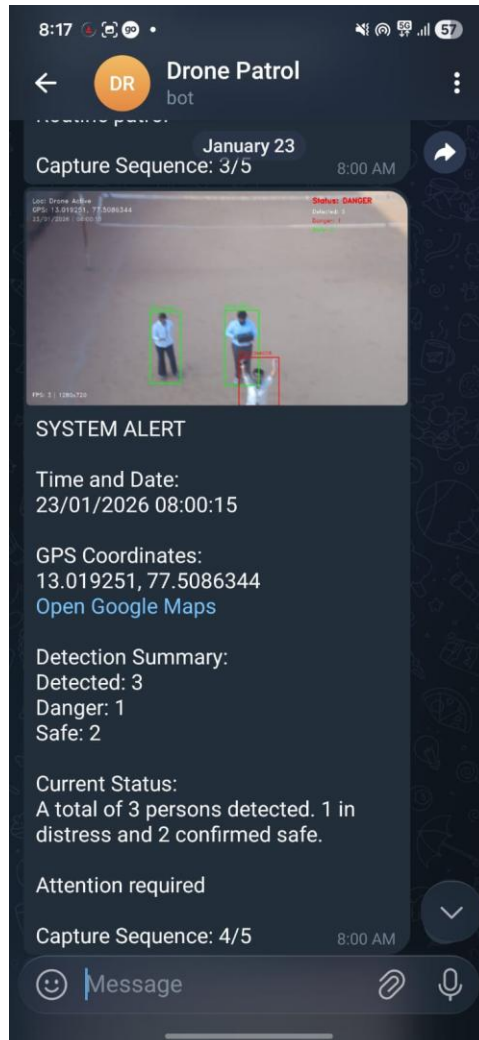
Alerting System

- **Notification Platform:** Upon distress, drone uses **Telegram Bot API** to notify operators instantly.
- **Data Sent:** Transmitted payload includes a snapshot of the person, GPS coordinates.
- **Rate Limiting:** To prevent spam, limit to e.g. 1 image per 15 frames ($\approx 0.5s$) and max 10 images per event.
- **Ground Station:** Operator receives alerts on mobile and can then direct rescue as needed.

Outcomes

- **Human Detection:** The system reliably detects people in real-time video (using YOLO) with high accuracy.
- **Gesture Alerts:** Accurately classifies “hands up” (distress) vs “hands down” (safe), triggering alerts when needed.
- **Real-Time Alerts:** Successfully sends image and location to ground, enabling faster rescue response.
- **Demonstration:** A prototype was assembled and tested indoors, showing stable flight and processing.





10.41.126.255 (WayVNC) - RealVNC Connect

drone17@raspberryp.. UAV TACTICAL MONI..

Trash

UAV TACTICAL MONITOR

loc: Drone Active
GPS: 13.0192304, 77.5086515
3/07/2026 | 07:58:44

Status: PATROL
Detected: 0
Danger: 0
Safe: 0

PS: 3 | 1280x720
51, y=288] - R:193 G:188 B:191

Mission Planner 1.3.83 build 1.3.9384.38258 ArduCopter V4.6.3 (92b0cd78) on Pixhawk...

DATA PLAN SETUP CONFIG SIMULATION HELP

165 S 195 2213 SW 240 255 W

07:58:44 99%

DISARMED SAFE

PreArm: Hardware safety switch

AS 1.2m/s GS 1.0m/s Stabilize 0m>0

Not Ready to Arm

Bat1 0.00v 0.0 A 0% EKF Vibe GPS: 3D dgps

Quick Actions Messages PreFlight Gauges Transponder St

Altitude (m)	GroundSpeed (m/s)
0.33	0.98
Dist to WP (m)	Yaw (deg)
0.00	213.20
Vertical Speed (m/s)	DistToMAV
0.66	3.38

Zoom in to see Airspace

hdop: 0.9

Sats: 9 Current Heading

©2026 Google - Map data ©2026

GEO 13.0192396

Future Scope

1. Autonomous Thermal-Based Human Detection

Integrate a thermal camera to enable autonomous detection of humans in low visibility conditions such as night, smoke, fog, or rubble. This will improve rescue efficiency during real disaster scenarios where RGB cameras fail.

2. Fully Autonomous Navigation & Area Scanning

Enhance the system with GPS-based waypoint navigation and obstacle avoidance, allowing the drone to autonomously patrol disaster zones without continuous manual control.

3. Multi-Modal Sensor Fusion

Combine RGB camera, thermal imaging, GPS, and IMU data to improve detection accuracy, reduce false alerts, and provide richer situational awareness to rescue teams.

4. Cloud & Multi-Drone Coordination

Extend the system to support cloud-based data storage and multi-drone coordination, enabling real-time mapping and collaborative search operations over large disaster-affected areas.

References

1. Liu C., Szirányi T. “Real-Time Human Detection and Gesture Recognition for On-Board UAV Rescue,” *Sensors*, 21(6):2180 (2021).
2. Taylor B. et al. “Enhancing Drone Navigation and Control: Gesture-Based Piloting, Obstacle Avoidance, and 3D Trajectory Mapping,” *Applied Sci.*, 15(13):7340 (2025).
3. Gaitan N.C., Batinas B.I., Ursu C. “AI-Enhanced Rescue Drone with Multi-Modal Vision and Cognitive Agentic Architecture,” *AI*, 6(10):272 (2025)

THANK YOU